

Quantum-Inspired Secure Authentication

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1 Description

With recent advancements in technologies across AI & machine learning as well as quantum computing, there have been proposed theoretical proof of concepts, showcasing numerous threats that specifically target areas as that of cybersecurity [1][2][3]. Quantum Computers could essentially harness large computational power proving advantageous over a classical computer; this would mean our current cryptographic / encryption algorithms which are used day to day in data transmission or authentication need to be "quantum-proof" [4][5].

We propose an authentication framework integrating behavioral biometrics, quantum inspired probabilistic identity modeling, and post quantum cryptographic security for future resistant protection. Specifically, we combine behavioral patterns of a user which include Keystroke Dynamics, Mouse Dynamics, Touch / Tap Patterns, Device Interaction ,Pattern Features (sequence, n-grams), Movement Features (trajectory, speed, angle), Session Context etc. We then represent user behavior as probabilistic state vectors inspired by quantum superposition concepts; onto which lattice based cryptography and hash based signatures (secure identity vectors) are applied to safeguard against quantum attacks. The system then continuously monitors the user behavior using ML based anomaly detection which provides real time alerts for any unusual behavior.

2 Previous Work

Behavioral biometrics and continuous authentication have been widely studied as a way to overcome the weaknesses of passwords and traditional biometrics in mobile, IoT, and high risk environments. Large surveys classify keystroke, mouse, touch, motion, and other behavioral signals, review feature extraction pipelines, and compare ML models for continuous authentication, while also highlighting adversarial vulnerabilities and drift issues[6]. Recent systems move from one time login to continuous, session long monitoring, including in smartphones and healthcare. In healthcare, post quantum continuous authentication has been proposed using behavioral biometrics and vector similarity search to avoid the weaknesses of one-time, RSA / ECC based login in the presence of Shor's algorithm[7][8].

At the cryptography layer, post quantum secure biometric schemes have used Classic McEliece or CKKS homomorphic encryption to protect facial or multimodal templates while preserving matching accuracy and enabling real time operation [9][10]. Quantum safe MFA frameworks for banking already integrate lattice based PQC with behavioral analytics and deep learning to achieve quantum resistant, low latency authentication [11].

However, existing post quantum biometric systems typically secure templates cryptographically rather than changing the biometric representation itself [9]. Surveys of behavioral biometrics and continuous authentication emphasize traditional time domain features and standard ML models, and mainly treat quantum computing as a cryptographic threat rather than a modeling opportunity [6][12]. The quantum inspired authentication framework remains largely conceptual, without a concrete encoding pipeline such as FFT based amplitude encoding or explicit use of quantum fidelity as the matching score.

3 Gap and Problem Statement

Across this prior work, key pieces exist but they remain largely disjoint: behavioral biometrics with powerful AI models, post quantum cryptography and QKD for transport security, and early quantum inspired identity modeling. Yet there is no concrete system that ;encodes behavioral biometrics into a quantum inspired state vector (e.g., normalized FFT spectrum), uses "quantum style" similarity metrics such as fidelity for continuous authentication, and secures these identity signatures end-to-end with standardized post quantum cryptography and, optionally, QKD. Existing continuous authentication and behavioral surveys point out vulnerabilities to spoofing, adversarial attacks, and drift (by Stylios et. al.) [6], while PQ biometric frameworks focus on template protection rather than on making the behavioral representation itself harder to copy or mimic [9][10]. The problem this project explicitly targets is designing and evaluating a quantum inspired secure authentication system that:

1. Captures rich behavioral signals and converts them via FFT into normalized spectral signatures.
2. Interprets these as amplitude encoded identity states and performs matching via quantum fidelity combined with an ML anomaly detector for continuous session monitoring.
3. Stores and transports the resulting templates using standardized PQC and compatible quantum safe algorithms (hybrid PQC and QKD).

Hence closing the current gap between behavioral AI, quantum cryptography, and quantum inspired modeling in a single, implementable architecture.

4 STAR

Situation

Traditional authentication (passwords, OTPs, static biometrics) is increasingly vulnerable to phishing, credential stuffing, spoofing, and future quantum attacks. The passwords and conventional biometrics can be compromised and that one time login models leave a “session security gap” where hijacking and privilege escalation can occur. At the same time, advancing quantum computing threatens classical cryptography such as RSA / ECC.

Task

The project’s objective is to design and implement a quantum resistant authentication framework that integrates:

1. Behavioral biometrics for richer user identity signals.
2. Quantum inspired probabilistic identity modeling.
3. Post quantum cryptographic protection of identity data and reaches high accuracy with low false acceptance/rejection rates.

Action

The system is built as a multi-layer, modular architecture:

1. Behavioral Biometric Capture: Collects keystroke dynamics, mouse movement patterns, and touch interaction timing at the frontend to build unique user signatures.
2. Feature Extractor and Identity Engine: A feature extractor analyzes behavioral patterns, and an identity engine represents user behavior as probabilistic state vectors inspired by quantum superposition concepts.
3. Post Quantum Security Layer: A security layer applies lattice based cryptography and hash based signatures to secure identity vectors against quantum attacks.
4. Decision Engine and Continuous Authentication: An ML based decision engine performs authentication and anomaly detection for real time session monitoring, triggering alerts or re-auth when deviations are detected.
5. Secure Storage: Identity templates are stored in an encrypted database as part of the secure storage module.

Result

Result will be added after project creation and thorough evaluation.

5 Use Cases

The framework targets high security, high impact domains:

1. Defense and Intelligence Systems: Continuous, quantum resistant access control for classified systems and command and control infrastructure where session hijacking or credential theft could have strategic consequences.
2. Critical Infrastructure Control: Protection for control consoles managing power grids, water systems, and transportation networks, ensuring only authenticated operators remain in control throughout a session.
3. Healthcare Information Systems: Securing EHR and clinical workstations with continuous monitoring to prevent session hijacking in hospitals and clinics, while resisting future quantum decryption threats to stored identity data.
4. Financial Services and Trading Platforms: Quantum resistant authentication for online banking, payment systems, and high frequency trading platforms, where behavioral biometrics plus Post Quantum encryption reduce fraud and “harvest-now, decrypt-later” risks.

6 References

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